Competitive Exam Practice 1: Introduction to Operating Systems

GATE (Computer Science) Pattern

Auqib Hamid Lone

Department of Computer Science & Engineering, GCET Ganderbal

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Concepts That Click

Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type. *All questions in this set are original, written in the GATE pattern — the repo holds no verifiable GATE OS past-year source.*

Q1 *[Original, GATE-pattern]* **[MCQ, 1 mark]** Which of the following is **NOT** a function performed by the operating-system kernel?

(A) Scheduling the CPU among ready processes

(B) Translating a C source file into assembly code

(C) Allocating and protecting physical memory frames

(D) Dispatching I/O requests to device drivers

Q2 *[Original, GATE-pattern]* **[MCQ, 1 mark]** A user process attempts to execute an instruction that modifies the base and limit registers. The processor raises a trap and does not perform the modification. The most direct hardware reason is that:

(A) the instruction is illegal in all processors

(B) base and limit registers are read-only in hardware

(C) the instruction is privileged and the process is running in user mode

(D) the process does not hold the file-system lock

Q3 *[Original, GATE-pattern]* **[NAT, 2 marks]** A system calls the kernel through the *parameter-block* method. Each system call has a fixed entry overhead of 3 μs (mode switch, dispatch, return) and, in addition, 0.4 μs per 8 bytes of parameter block copied into the kernel. A call passes a parameter block of 40 bytes. The total time (in μs) spent inside the kernel for this call is _____.

Q4 *[Original, GATE-pattern]* **[MCQ, 2 marks]** Consider a microkernel-based OS and a monolithic OS running on identical hardware. A device driver contains a bug that corrupts memory. Which statement is **TRUE**?

(A) Both architectures isolate the driver, so neither is affected

(B) In the monolithic OS the corruption propagates into kernel state; in the microkernel the driver runs as a user-space server that can be restarted

(C) In the microkernel the corruption propagates into kernel state; in the monolithic OS the driver is isolated

(D) Both architectures crash, since drivers always run in kernel mode

Q5 *[Original, GATE-pattern]* [**MCQ, 1 mark**] Which of the following statements about system calls is **TRUE**?

(A) A library call in the POSIX API always corresponds to exactly one system call

(B) A system call executes entirely in user mode

(C) A system call is the controlled mechanism by which a program requests a privileged kernel service

(D) A system call cannot transfer control to a device driver

Q6 *[Original, GATE-pattern]* [**NAT, 2 marks**] A utility reads a file of 8 MB using a 4 KB buffer and writes it to another file, issuing exactly one `open`, one `read` and one `write` per buffer, and one `close`. The total number of system calls issued is _____.

Q7 *[Original, GATE-pattern]* [**MCQ, 1 mark**] The primary objective of *multiprogramming* — as distinct from time-sharing — is to:

(A) minimize the response time of each interactive user

(B) maximize CPU utilization by overlapping I/O waits with computation

(C) guarantee hard real-time deadlines

(D) eliminate the need for a scheduler

Q8 *[Original, GATE-pattern]* [**MCQ, 1 mark**] The statement "the operating system presents a *file* to the programmer instead of an array of disk blocks" is best described as illustrating the OS as:

(A) a control program

(B) a resource allocator

(C) an extended machine (abstraction layer)

(D) an accounting program

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